



Dynamic Pricing and Matching for Two-Sided Queues

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Abstract

Motivated by diverse applications in sharing economy and online marketplaces, we consider optimal pricing and matching control in a two-sided queueing system. We assume that heterogeneous customers and servers arrive to the system with price-dependent arrival rates. The compatibility between servers and customers is specified by a bipartite graph. Once a pair of customer and server are matched, they depart from the system instantaneously. The objective is to maximize the long-run average profits of the system while minimizing average waiting time. We first propose a static pricing and max-weight matching policy, which achieves $O(\sqrt{\eta})$ optimality rate when all of the arrival rates are scaled by η . We further show that a dynamic pricing and modified max-weight matching policy achieves an improved $O(\eta^{1/3})$ optimality rate. In addition, we propose a constraint generation algorithm that solves value function approximation of the MDP and demonstrate strong numerical performance of this algorithm.

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1 Introduction

A two-sided queueing system is one where customers and servers both arrive and then wait in queues until they are matched. Many emerging applications and rapidly growing marketplaces can be modeled as such systems. Some examples include Uber and Lyft where passengers are matched with drivers, Grubhub where customers are matched with meal delivery couriers, and crowdsourcing platforms where requesters are matched with contributors. Most of these platforms use both dynamic pricing and dynamic matching as levers to facilitate market profitability and efficiency.

In this paper, we consider a canonical model of a two-sided queueing systems with n types of servers and m types of customers. Each type of customer is allowed to be matched with only a subset of server types. For example, in the case of ride hailing system, the type of servers (cars) and passengers (customers) will be determined by their geographical locations. Customers and servers which are nearby will be compatible to be matched to each other.

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At each point in time, the system operator sets a price for each type of customer and server. Then, customers who are willing to pay and servers who are willing to serve for the quoted price enter the system. They each wait in separate queues, until they are matched to a compatible counterpart. Once a pair is matched to each other, they will leave the system immediately in order to complete the service. The system operator earns a profit, which is equal to the difference between the price charged to the customer and the price quoted to the server.

We formulate the above system as a Markov decision process (MDP). The operator can vary the price for each type of customer and server, as well as decide when to match and which customer-server pair to match. The objective is to maximize the profits obtained by the system operator, while accounting for the queuing delay experienced by the customers and servers. We are especially interested in the behavior of systems with large traffic, as all of the arrival rates are scaled by a factor $\eta \rightarrow \infty$. Under this scaling regime, any policy that is within $o(\eta)$ of the optimal objective is asymptotically optimal.

The main challenge in this problem is the *curse of dimensionality* in solving the MDP. As the number of customer and server types increases, the dimension of the state space increases exponentially, even when the buffer size of the queue for each type is bounded by a constant. Thus, it is intractable to solve the exact MDP for large scale systems, which are often encountered in practice. In this paper, we propose approximation technique to obtain near optimal solutions for the MDP efficiently. In addition, we propose dynamic pricing and matching policies that are asymptotically optimal. We model a bipartite two-sided queueing network with stochastic arrivals as a Markov decision process. In some special cases, we are able to show structural properties of the optimal dynamic pricing policy. We then analyze a fluid model. The revenue obtained by solving the fluid model is an upper bound on the achievable revenue under any policy. We propose a fluid pricing and max-weight matching policy, and show that the revenue loss from the fluid revenue is $O(\sqrt{\eta})$. Thus, it is asymptotically optimal. We also propose a two price and modified max-weight matching policy, and show that the revenue loss from the fluid revenue is reduced to $O(\eta^{1/3})$. Finally, we present an approximation technique and a constraint generation algorithm to solve the MDP efficiently. The approximation sometimes leads to a closed-form expression of the pricing policy.

2 Model

We model the multiple types of customers and servers by a bipartite graph $G(N_1 \cup N_2, E)$, where N_1 is the set of server types with $|N_1| = n$ and N_2 is the set of customers type with $|N_2| = m$. Here, E are the set of edges in the bipartite graph between customer and server types which are compatible with each other. In other words, $(i, j) \in E$ if type j customer can be served by type i server. Each node in the bipartite graph is a queue of customers or servers waiting to be matched with any one of the compatible counterpart.

Our convention is to refer the incoming demand as customers and incoming supply as server.

We assume that customers and servers arrive according to Poisson processes with rate $\lambda \in \mathbb{R}_+^m$ and $\mu \in \mathbb{R}_+^n$ respectively. For each customer type $j \in [m]$, there exists a demand curve $F_j : \lambda_j \rightarrow p_j^{(2)}$, such that if the system operator sets a price $p_j^{(2)}$, the resulting arrival rate is λ_j . Similarly, for each server type $i \in [n]$, the system operator can set a price $p_i^{(1)}$, which results in a server arrival rate μ_i determined by the respective supply curve $G_i : \mu_i \rightarrow p_i^{(1)}$. We make mild assumptions that the supply and demand curve are monotonic, continuously twice differentiable, has a compact domain. We also assume that the cost function $G_i(\mu_i)\mu_i$ is convex and the revenue function $F_j(\lambda_j)\lambda_j$ is concave. When a pair of customer and server is matched by the system, they both depart from the system. Note that since not all customer and server types are compatible, the system operator may have incentive for *batching*, i.e., holding some customers and servers in queue in order to achieve more favorable matches in future.

We formulate the problem as an infinite horizon Markov decision process. The state is the number of customers and servers waiting in the system to be processed, which is denoted by $\mathbf{q} \in \mathbb{Z}_+^{m+n}$. Our objective is to find an optimal dynamic pricing and matching policy under which the long-run average profit is maximized. The profit consists of revenues collected from customers, payments to the servers, and a penalty term proportional to the number of customers and servers waiting in queues.

3 Static Price versus Two Price Policy

Fluid Model We consider a fluid approximation of the system with deterministic customer arrival rates λ and server arrival rates μ .

$$\begin{aligned} \max \quad & \langle F(\lambda), \lambda \rangle - \langle G(\mu), \mu \rangle \\ \text{s.t.} \quad & \lambda_j = \sum_i \chi_{ij}, \forall j \in [m], \mu_i = \sum_j \chi_{ij}, \forall i \in [n], \\ & \chi \in \mathbb{R}_+^{n \times m}, \chi_{ij} = 0, \forall (i, j) \notin E. \end{aligned} \quad (1)$$

The solution to the above problem is denoted by λ^* and μ^* . In addition, the optimal fluid revenue is defined as $R_*^\eta = \langle F(\lambda^*), \lambda^* \rangle - \langle G(\mu^*), \mu^* \rangle$. It can be shown R_*^η is an upper bound of the optimal profits for the original stochastic problem. Next, we state the asymptotic regime under which we analyze our algorithm.

Definition 3.1 (Asymptotic Regime). Consider a family of CTMCs parametrized by η such that for the η^{th} CTMC, the demand and supply curve satisfy $F^\eta(\eta\lambda) = F(\lambda)$ for all $\lambda \in \mathbb{R}_+^m$ and $G^\eta(\eta\mu) = G(\mu)$ for all $\mu \in \mathbb{R}_+^n$. The asymptotic regime is defined as $\eta \rightarrow \infty$.

We use the following to measure the performance of algorithms.

Definition 3.2 (Revenue Loss L^η). It is defined as the expected value of the difference of the fluid revenue R_*^η and the revenue obtained (including the penalty incurred due to waiting) when the CTMC is operating under the given policy. Mathematically,

$$L^\eta = R_*^\eta - (R^\eta - s \langle \mathbf{1}_{m+n}, \mathbb{E}[\mathbf{q}] \rangle),$$

where s (constant) is the penalty coefficient and R^η is the expected revenue under the given policy.

3.1 Static Price Policy

Pricing Policy We use fluid optimal rates λ^* and μ^* if queue length is less than a threshold $\mathbf{q}_{max}^\eta$ and reject all the arrivals otherwise.

Max-Weight Matching Policy The system operator is allowed to match every time the system state changes. Whenever there is an arrival, if any of the compatible counterparts of the arrival is waiting in the queue, then we match it to the type with most number of customers/servers waiting in that queue. On the other hand, if all the compatible counterparts' queues are empty, then we can't match it to anyone and that arrival will then wait in the queue until in the future, some other arrival is matched to it.

THEOREM 3.3. *For the family of multiple link two-sided queues with a given compatibility graph $G(N_1 \cup N_2, E)$ parametrized by η operating under the Fluid pricing policy and max-weight matching policy, the revenue loss L^η due to stochasticity is $O(\sqrt{\eta})$ when $\mathbf{q}_{max}^\eta = \gamma\sqrt{\eta}$ for any vector of constants γ .*

3.2 Two Price Policy

To get better performance than static price, we adjust price when the queue length exceeds some threshold. More specifically, we introduce additional parameters θ and ϕ which governs the arrival rates of the customers and servers respectively when the queue length is greater than a certain threshold $\mathbf{q}_{tp}$. We have

$$\lambda_j^\eta = \begin{cases} \eta\lambda_j^* & \text{if } q_j^{(2)} \leq q_{tpj}^{\eta(2)}, \\ \eta\lambda_j^* - \theta_j^\eta & \text{otherwise} \end{cases}, \forall j \in [m], \quad (2)$$

$$\mu_i^\eta = \begin{cases} \eta\mu_i^* & \text{if } q_i^{(1)} \leq q_{tpi}^{\eta(1)}, \\ \eta\mu_i^* - \phi_i^\eta & \text{otherwise} \end{cases}, \forall i \in [n]. \quad (3)$$

To expound, we set the prices as a function of the queue lengths such that the arrival rate of customers and servers follows (2) and (3) respectively. Here, $\mathbf{q}_{tp}^\eta$, θ^η and ϕ^η are parameters which can be chosen by the operator.

The matching policy we use is max-weight policy but we never match using some of the edges in the graph. The technical details for the same can be found in the full paper [1, Section 5].

THEOREM 3.4. *For a family of multiple link two-sided queues with a given compatibility graph $G(N_1 \cup N_2, E)$ parametrized by η operating under the two price policy and modified max-weight matching policy, the revenue loss $L^\eta = O(\eta^{1/3})$ for $\mathbf{q}_{tp}^\eta = \Theta(\eta^{1/3})$, $\theta^\eta = \Theta(\eta^{2/3})$ and $\phi^\eta = \Theta(\eta^{2/3})$.*

The key observation is that there is a trade off between the revenue obtained and the expected queue length of the system. The idea is to tune $\mathbf{q}_{tp}^\eta$, θ^η and ϕ^η such that $\mathbb{E}[\langle \mathbf{1}_{m+n}, \mathbf{q} \rangle] \sim \eta^{1/3}$ and $R_*^\eta - R^\eta \sim \eta^{1/3}$. By Theorem 3.3 and Theorem 3.4, even a simple two-price dynamic pricing policy has a much better performance than a static (fluid) pricing policy. It is also practically easy to implement the two price policy.

References

- [1] Sushil Mahavir Varma, Pornpawee Bumpensanti, Siva Theja Maguluri, and He Wang. 2019. Dynamic Pricing and Matching for Two-Sided Queues. *arXiv preprint arXiv:1911.02213* (2019).